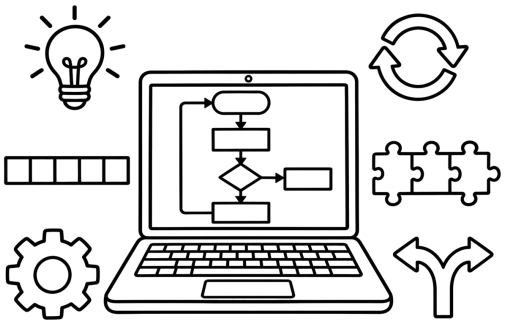
ICT

INFORMATION AND COMMUNICATION TECHNOLOGY

GRADE 11



UNIT 1 - PART 1
Programming

Student Workbook

"The King of ICT"

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STUDENT DETAILS

Name: _____

School: _____

Institution: _____

Contact: _____

BEFORE WE START

What This Part Is About

Unit 1 Programming - Part 1 of 2

AT THE END OF THIS PART YOU WILL BE ABLE TO

- [ ] Analyze a problem into its input, process and output.
- [ ] Find alternative solutions and choose the most suitable one.
- [ ] Write algorithms with sequence, selection and repetition.
- [ ] Draw flowcharts with the standard symbols of Table 1.1.
- [ ] Write pseudo codes and convert flowcharts to pseudo codes.
- [ ] Use identifiers, reserved words, data types, variables and constants.
- [ ] Evaluate expressions with arithmetic, comparison and logical operators.

HOW TO USE THIS BOOK

Read each chapter with your teacher, do the practical tasks and then write the exercises. The **WRITING SPACES** are for the notes your teacher gives you in class - write them neatly: they are your revision notes for the O/L examination.



THE WORDS YOU WILL LEARN

Input	Process	Output	Alternative solution
Solution space	Algorithm	Control structure	Sequence
Selection	Repetition	Flowchart	Decision
Connector	Pseudo code	Identifier	Reserved word
Data type	Variable	Constant	Operator
DIV	MOD	Precedence	Assignment

"Every program starts on paper: input, process, output. Plan first, then type."

- THE KING OF ICT

WRITING SPACE 1 - What is a program?

write what your teacher tells you

CHAPTER 1

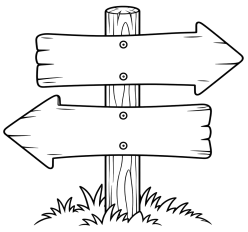
Problem Solving and Algorithms

THE SCHOOL BUS BROKE DOWN!

"Our school bus broke down on the way this morning!"

"How did you reach school?" "By a CTB bus. There was more than one way."

A problem has an **input**, a **process** and an **output**, and it may have **alternative solutions**. We choose the best one and write its steps as an **algorithm**.



1.1 Analyzing a problem

The raw materials used to solve a problem are the **input**. The result obtained after solving it is the **output**. Converting the input to the output is the **process**. A process takes place **step by step**, and the order of the steps is very important. When we analyze a problem, we identify the input, the process and the output separately.

Problem	Input	Process	Output
Problem 2: Making a cup of tea	Tea leaves, sugar, hot water	(1) Putting tea leaves in the strainer (2) Pouring hot water to the cup through the strainer (3) Adding some sugar (4) Stirring it well with a spoon (5) Testing for taste (6) If the taste is not satisfactory, go to step 3 and repeat steps 4 and 5	A cup of tea
Problem 4: Adding two numbers	Two numbers	Adding the two numbers	Total
Problem 5: Finding the area of a rectangle	Length and width of the rectangle	Area = Length × Width	Area

WRITING SPACE 2 - Input, process and output — write what your teacher tells you

WRITING SPACE 3 - Alternative solutions — write what your teacher tells you

Alternative solutions

If there is more than one solution to a problem, they are called **alternative solutions** (the school bus: another bus, a CTB bus, a short-cut ...). All the solutions pertaining to a problem are the **solution space**. In programming, too, we identify the solutions and select an appropriate one; then we can create a short, simple program.

EXAMPLE 1: THE PERIMETER

Input: length and width.

1. Perimeter = length + width + length + width
2. Perimeter = length × 2 + width × 2
3. Perimeter = (length + width) × 2

Only addition: the 1st. Multiplication too: the **3rd** - the fewest additions and multiplications.

EXAMPLE 2: PASS OR FAIL

Fail if the ICT mark is less than 35, pass if 35 or above.

Solution 1. If the mark is less than 35: Result = Fail. If not: Result = Pass.
Solution 2. If the mark is 35 or more than 35: Result = Pass. If not: Result = Fail.

EXAMPLE 3: LARGER NUMBER

Two numbers n1 and n2 (Problem 6).

Solution 1. If n1 is larger than n2, the larger number is n1. If n2 is larger than n1, the larger number is n2.
Solution 2. Subtract n2 from n1. If the result is more than 0, n1 is the larger number. If the result is less than 0, n2 is the larger number.

1.2 Problem solving using algorithms

An **algorithm** is a method to show the steps in solving a problem: a **step-by-step procedure** for solving it. It presents a way to solve the problem with a **plan**.

Some steps of an algorithm must be followed in a **strict sequential order**. For some other steps, changing the order does not affect the process, and the output is the same.

EXAMPLE: MEASURING 500 g OF SUGAR

(1) Putting sugar into a bag
(2) Placing the bag on the scale and getting the reading
(3) If the weight of the sugar is less than 500 g, add sugar till it weighs 500 g
(4) If the weight of the sugar is more than 500 g, remove sugar till it weighs 500 g
(5) Remove the bag of sugar from the scale when the weight is 500 g

1.2.1 Control structures

Three types of control structures are used in algorithms: **sequence**, **selection** and **repetition**.

Control structure	What it is	Examples (textbook pages 7 - 9)
Sequence	The steps from the beginning to the end are carried out in a **strict order**.	Climbing up or down a staircase step by step. A student admitted to Grade 1 continues studies till Grade 13.
Selection	Step(s) are executed depending on whether a **condition** is satisfied or not: one choice if it is satisfied, the other if it is not.	Grade 1: if a child is below 5 years as at 31st January, the child cannot be admitted; if not, the child can be admitted. Passing a subject: 35 or more is a Pass. Buying a book.
Repetition	One or several steps are **repeated until a condition is satisfied**.	Marking the attendance register till the last name is called. Counting the words of a paragraph.

WRITING SPACE 4 - Three control structures write what your teacher tells you

DO IT - PRACTICAL 1 (ACTIVITIES, PAGES 3 - 9)

Problems, algorithms and control structures. Work in pairs and write your answers in your exercise book.

1	Identify the **input**, the **process** and the **output** of dividing 100 toffees among 20 people, and of making a kite (page 3).
2	A primary school has 183 students. Develop an **algorithm** to divide them into three houses: Olu, Nelum and Manel (page 6).
3	Write down two incidents which consist of **sequences** and three which consist of **selection** (pages 7 - 8).
4	A citizen gets the right to vote after completing the age of 18. Complete: If the age is ..... (more / less) than 18, the vote ..... (can / cannot) be cast; if not, the vote ..... (can / cannot) be cast (page 8).
5	Write down, as numbered steps, two scenarios that comprise **repetition** (page 9).

A Input, process and output

9 marks

Write the input, the process and the output of each problem (1 mark each).

Problem	Input	Process	Output
Problem 3: Dividing 40 page and 80 page books from a parcel between Sanduni and Anupama			
Problem 6: Finding the larger number between two numbers			
Problem 7: Finding whether a number is odd or even			

WRITING SPACE 5 - Examples of selection write what your teacher tells you

B Sequence, selection or repetition?

8 marks

Tick the control structure of each example.

Example	Sequence	Selection	Repetition
Climbing up a staircase, step by step.	☐	☐	☐
If the mark is 35 or more it is a Pass; if not it is a Fail.	☐	☐	☐
Calling the names on the register till the last name is called.	☐	☐	☐
A student admitted to Grade 1 continues studies till Grade 13.	☐	☐	☐
If you have money equal to or more than the price of the book, you can buy it.	☐	☐	☐
Reading the next word and adding 1, till the end of the paragraph.	☐	☐	☐
A child below 5 years as at 31st January cannot be admitted to Grade 1; if not, the child can.	☐	☐	☐
Adding sugar to the bag till it weighs 500 g.	☐	☐	☐

C Put the algorithm in order

8 marks

Posting a letter: write 1 - 6 in the boxes, then answer.

a.	☐	Sticking the stamp
b.	☐	Writing the letter
c.	☐	Posting the letter
d.	☐	Inserting the letter in an envelope
e.	☐	Writing the address
f.	☐	Folding the letter

1. Which two steps can be interchanged without changing the output? Why? (2 marks)

D True or false

6 marks

Write T (true) or F (false) in the box.

1.	☐	When analyzing a problem, the input, process and output are identified separately.
2.	☐	A process takes place step by step, but the order of the steps is never important.
3.	☐	In the solution "Perimeter = length + width + length + width" there is no multiplication.
4.	☐	In an algorithm, the order of every step can be changed without changing the output.
5.	☐	There are three types of control structures: sequence, selection and repetition.
6.	☐	In selection, both choices are carried out one after the other.

CHAPTER 2

Flowcharts and Pseudo Codes

? A PLAN BEFORE THE PROGRAM

"Can I start typing the program now?"

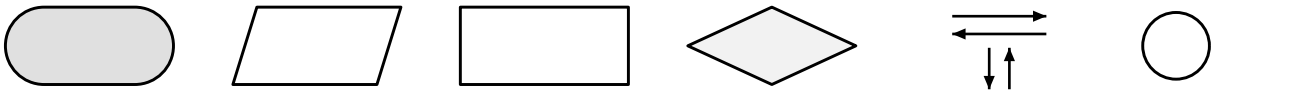
"First draw the plan: a flowchart or a pseudo code. Then the program is easy!"

Flowcharts and **pseudo codes** are tools to present an algorithm, so that it is understood better (1.3).



1.3.1 Flowcharts

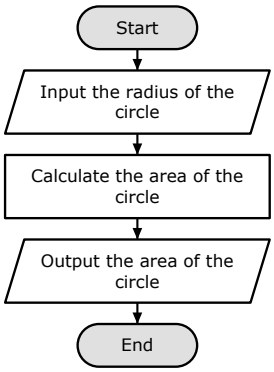
Flowcharts show how an algorithm is built step by step, in pictures. The symbols of Table 1.1 are used to indicate the different functions.



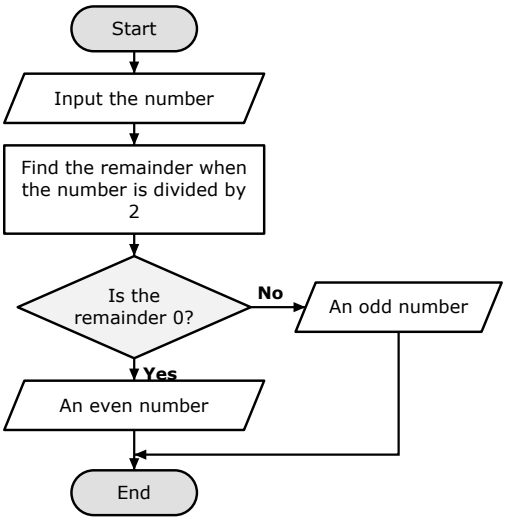
Start or end	Input or output	Process	Decision	Flow direction	Connector

Table 1.1 - the flowchart symbols

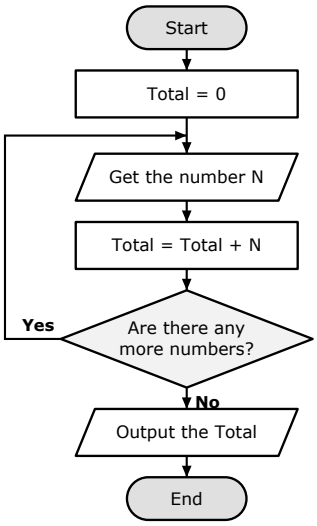
Sequence: the steps from the beginning to the end are executed in order. **Selection**: the flow of direction depends on a condition being satisfied or not. **Repetition**: steps are repeated till a condition is satisfied.



E.g. 1: the area of a circle



E.g.: odd or even



E.g.: the total of some numbers

"Draw the flowchart first. A good plan makes a short program."

- THE KING OF ICT

WRITING SPACE 6 - Flowchart symbols write what your teacher tells you

DO IT - PRACTICAL 2 (ACTIVITY, PAGE 12)

Draw flowcharts to solve these problems. Use the symbols of Table 1.1.

1	Find the perimeter and the area of a rectangle.
2	A company adds Rs. 5000 to the basic salary of its employees. Calculate the new salary.
3	Postal fare: the standard fare for a letter equal to or less than the standard weight; an additional fare if it is heavier.
4	Indicate the first 12 multiples of 7.
5	The examples of selection in 1.2.1: Grade 1 admission, passing a subject, buying a book. Then convert two of your flowcharts to pseudo codes (1.3.3).

1.3.2 Pseudo codes

When an algorithm is presented in **simple English terms**, it is called a **pseudo code**. Pseudo codes are independent of a computer language and can be converted to the instructions of any programming language, so they make programming easier.

Pseudo-code term	Used to ...
BEGIN	To indicate a beginning
END	To indicate an end
INPUT, READ, GET	To indicate an input
OUTPUT, DISPLAY, SHOW	To show an output
PROCESS, CALCULATE	To indicate a process
IF ... THEN ... ELSE ... ENDIF	Used to indicate a selection
FOR - DO **WHILE - ENDWHILE** **REPEAT - UNTIL**	Used to indicate a repetition

E.g. 1 - Finding the area of a circle

```
BEGIN
    INPUT Radius
    CALCULATE Area = 22/7 × Radius × Radius
    DISPLAY Area
END.
```

E.g. 2 - Finding whether a number is odd or even

```
BEGIN
    READ number as N
    CALCULATE Remainder after number
              is divided by 2
    IF Remainder = 0 THEN
        DISPLAY "Even number"
    ELSE
        DISPLAY "Odd number"
    ENDIF
END.
```

"Follow the arrows step by step, like the computer does."

- THE KING OF ICT

WRITING SPACE 7 - Pseudo code

write what your teacher tells you

E.g. 3 - The total of some numbers

```
BEGIN
    Total = 0
    REPEAT
        READ Number as N
        CALCULATE Total = Total + N
    UNTIL numbers are over
    DISPLAY Total
END.
```

E.g. 4 - The total and the average of 10 numbers

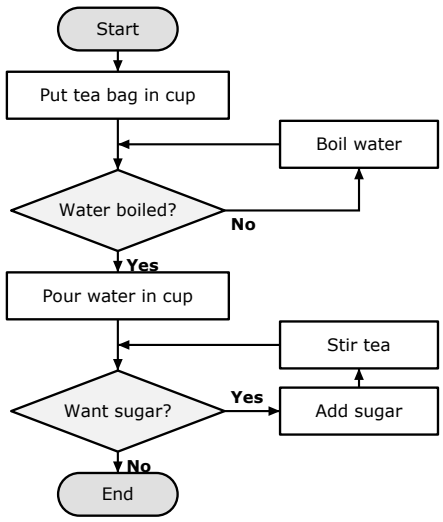
```
BEGIN
    Total = 0
    Average = 0
    n = 1
    WHILE n <= 10
        READ Number
        CALCULATE Total = Total + Number
        n = n + 1
    ENDWHILE
    CALCULATE Average = Total/(n - 1)
    DISPLAY Total, Average
END.
```

ABOUT E.g. 4

- **Total**, **Average** and **Number** are variables. **n** counts the repetitions (the number of times the loop is executed).
- **Total = 0** and **Average = 0** give the starting (initial) values; **n = 1** makes the starting value of n 1.
- **WHILE n <= 10**: the repetition occurs while n is 10 or less than 10. When n becomes 11, the condition is false and the repetition stops.
- Only READ Number, Total = Total + Number and n = n + 1 are repeated (they end at **ENDWHILE**).
- **Average = Total/(n - 1)**: n is 11 when the loop stops, so the total is divided by 10.

1.3.3 Converting flowcharts to pseudo codes

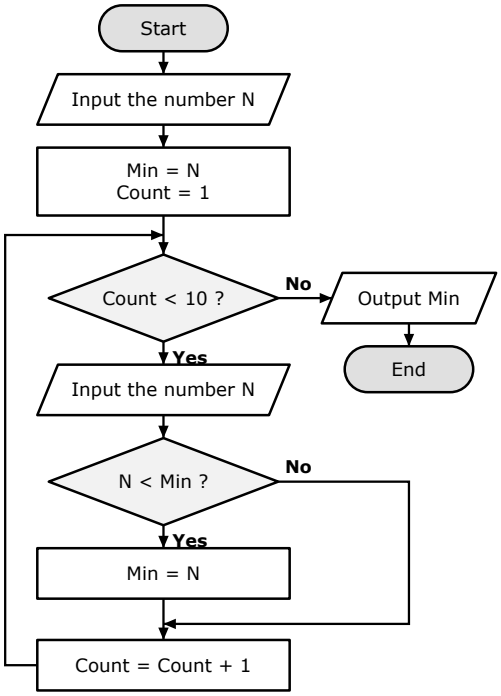
An algorithm can be presented as a flowchart and as a pseudo code: a loop becomes WHILE ... ENDWHILE and a decision with two branches becomes IF ... THEN ... ELSE ... ENDIF (E.g. 1, page 16).



Making a cup of tea

```
BEGIN
    Put tea bag in cup
    WHILE (not water boiled)
        Boil water
    ENDWHILE
    Pour water in cup
    WHILE (sugar needed)
        Add sugar
        Stir tea
    ENDWHILE
END
```

E.g. 2 - Finding the smallest number from 10 numbers

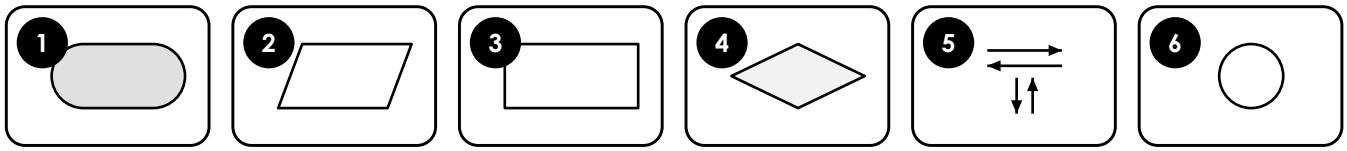



```
BEGIN
    INPUT Number as N
    Min = N
    Count = 1
    WHILE Count < 10
        INPUT Number as N
        IF N < Min Then
            Min = N
        ENDIF
        Count = Count + 1
    ENDWHILE
    PRINT Min
END.
```

A Name the symbols

6 marks

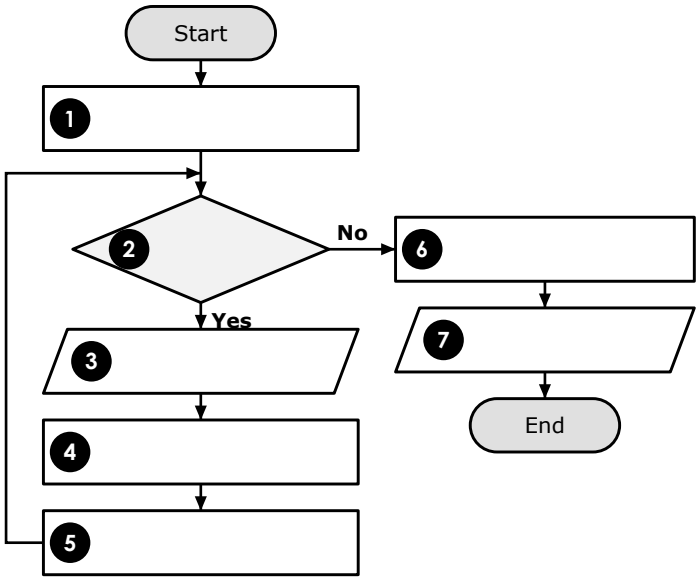
Write the function of each flowchart symbol.



B Complete the flowchart

7 marks

The flowchart of E.g. 4 (1.3.2): write the words in the numbered symbols.



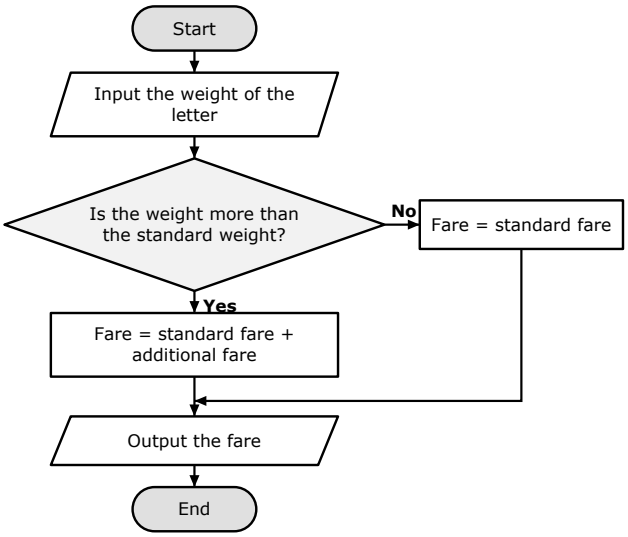
USE THESE

- Average = Total/(n - 1)
- n <= 10 ?
- Total = Total + Number
- Total = 0, Average = 0, n = 1
- DISPLAY Total, Average
- READ Number
- n = n + 1

C Convert the flowchart to pseudo code

6 marks

The postal fare of a letter (Activity, page 12).



PSEUDO CODE

D Follow the pseudo code

6 marks

E.g. 4 (1.3.2) reads these ten numbers. Answer.

The numbers: 4 7 2 9 5 6 3 8 1 5

1. What is the value of Total after the third number?

2. How many times is READ Number executed?

3. What is the value of n when the repetition stops?

4. What is the value of Total at the end?

5. What is the value of Average?

6. Why is Total divided by (n - 1) and not by n?

CHAPTER 3

Pascal Basics and Operators

? THE COMPUTER SAYS ERROR!

"I typed Total Marks as a name and the computer did not accept it!"

"An identifier has no spaces. Pascal has rules - let us learn them."

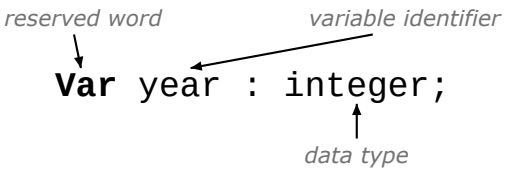
Pascal is a high-level programming language. A program has **identifiers**, **reserved words**, **data types**, **variables**, **constants** and **operators**.



1.4.1 Identifiers

An **identifier** is a term used to represent a variable, a constant or a program. The rules for an identifier:

- Reserved words of Pascal cannot be used as identifier names (BEGIN, END are not valid).
- An identifier should start with an English letter (A - Z, a - z).
- After the first letter: letters (a - z, A - Z), numbers (0 - 9) and the underscore (_). E.g. Student_name
- Pascal is not case sensitive: Art, art and ART are the same identifier.
- There should not be any space between words: Student Name is not valid.
- No special characters such as ~ ! @ # $ % ^ & * ( ) - + = { } [ ] : ; ' " < > ? , . / | \ - only the underscore (_) is valid.
- Meaningful identifiers make a program easily understood.



Valid	Not valid
Sum, SUM, Total_Nos	$75, Average Marks
Num1, FirstName, Last_Name	9A, Last-name

WRITING SPACE 8 - Identifiers — write what your teacher tells you

1.4.2 Reserved words

The **reserved words** are defined in the Pascal language, so they are not used as identifiers. Reserved words are different from language to language.

THE RESERVED WORDS OF PASCAL (TEXTBOOK PAGE 18)

and	do	if	nil	repeat	until
array	downto	implementation	not	set	uses
asm	else	in	object	shl	var
begin	end	inherited	of	shr	while
case	exports	inline	or	string	with
const	file	interface	packed	then	xor
constructor	for	label	procedure	to	
destructor	function	library	program	type	
div	goto	mod	record	unit	

"One wrong semicolon can stop a whole program. Be careful, be exact."

- THE KING OF ICT

1.4.3 Standard data types in Pascal

When a program is executed, the input and the output are stored in memory. The space needed for each is defined by its **data type**. **Important:** the values of Char and String are written inside single quotation marks: 'k', 'Sri Lanka'.

Data type	Values	E.g.
Integer	Plus or minus whole numbers	0, 46, -12
Real	Plus or minus decimal numbers	0.0, 25.68
Boolean	True or False	True, False
Char	Any character of the keyboard	'k', '#', '7'
String	Any sequence of characters	'ICT', 'programming', 'Sri Lanka'

Variable (name)	Value	Type
number	-123	integer
sum	456	integer
character	'B'	char
book	'Mathematics'	string

1.4.4 Variables and constants

A **variable** is an identifier which changes the value given to it while the program is being executed. In Pascal, **var** is used to declare variables. A variable has a name and stores a value of the declared type.

```
var count : integer;
var a,b : real;
var
    n1, n2 : integer;
    Avg : real;
    Pass : boolean;
    Character : char;
    Name,school : String;
```

The identifiers which do not change their values while the program is executed are **constants**. In Pascal, **const** is used to declare a constant.

```
Const max = 100;
Const pi = 22/7;
```

Observation: when a program is executed, a variable can take different values, but the value of a constant remains unchanged.

1.5 Operators

Operators are required to perform calculations and comparisons and to evaluate logical expressions. **1. Arithmetic (algebra) operators** and **2. comparison operators**: the result of a comparison is always a **Boolean** value, True or False.

Operator	Usage	Example	Result
+	Addition	6 + 3	9
-	Subtraction	7 - 5	2
*	Multiplication	2 * 5	10
/	Division	10/4	2.50
DIV	Division of round numbers	20 DIV 6	3
MOD	Remainder after division	20 MOD 6	2

Operator	Usage	Example	Result
>	Greater than	7 > 3	True
>=	Greater than or equal	8 >= 8	True
<	Less than	3 < 2	False
<=	Less than or equal	4 <= 6	True
=	Equal	3 = 1	False
< >	Not equal	2 < > 5	True

"Give your variables meaningful names. Your teacher, and you, will thank you."

- THE KING OF ICT

3. Logical operators combine two or more expressions. **AND** is true only when both expressions are true. **OR** is false only when both expressions are false. **NOT** changes true to false and false to true. E.g. (3 >= 2) AND (3 < > 3) is False; (3 >= 2) OR (3 < > 3) is True; NOT (5 = 5) is False.

First	Second	AND
False	False	False
False	True	False
True	False	False
True	True	True

First	Second	OR
False	False	False
False	True	True
True	False	True
True	True	True

Expression	NOT
False	True
True	False

3 ← **DIV**
6 | 20 18
2 ← **MOD**
20 DIV 6 = 3
20 MOD 6 = 2

Operator precedence and evaluating expressions

When Pascal expressions are evaluated, the operators with a higher priority are evaluated first (NOT is the highest; the comparisons are the lowest).

Priority	Operator
1	NOT
2	* / DIV MOD AND
3	+ - OR
4	= < > < <= > >=

Expression	Working	Result
5 + 14 MOD 4	5 + 2	7
3 + 7 DIV 2	3 + 3	6
16 / 4 * 2	4 * 2	8
NOT (8 MOD 2 > 5)	NOT (0 > 5) = NOT (False)	True
(4 >= 4) AND NOT (7 > 9)	True AND NOT (False) = True AND True	True

WRITING SPACE 9 - Operator precedence — write what your teacher tells you

The parts of a Pascal program

```
1  program addNum (input,output);                          1
2  var
3      num1,num2,total : integer;                          2
4      avg: real;
5  Begin                                                   3
6      Write('Welcome Pascal Programming');
7      {calculate total and average of two numbers}        4
8      Writeln('Enter First Number');
9      Read(num1);                                         5
10     Writeln('Enter Second Number');
11     Read(num2);
12     total := num1 + num2;                               6
13     avg := total/2;
14     writeln('Total is ', total);                        7
15     writeln('Average is ', avg);
16 End.                                                    8
```

The program addNum (textbook page 24)

1. The heading: the name of the program (addNum)
2. Declaring the variables
3. Beginning of the main program
4. A comment: { } or (* *)
5. Input: Read( )
6. The process (an assignment with :=)
7. Output: writeln( )
8. Ending of the main program (End.)

A semicolon (;) is used at the end of a statement. **:=** is the assignment operator. Comments: { ... } or (* ... *).

read( ) and readln( ) are for input (readln from a new row); write( ) and writeln( ) for output (writeln in a new line).

DO IT - PRACTICAL 3 (TEXTBOOK PAGES 24 - 25)

Type and run addNum in a free Pascal compiler (Free Pascal or Lazarus). **Ask your teacher first.**

1	Open the Free Pascal IDE (or Lazarus). Click **File, New**. Type the program **addNum** exactly as it is above and save it as addnum.pas.
2	Click **Compile, Compile** (Alt+F9). The compiler highlights any error: check the ; and the quotes.
3	Click **Run, Run** (Ctrl+F9). Type 12 and press Enter; type 8 and press Enter.
4	Press **Alt+F5** to see the output screen. Is the total 20? Is the average 10?
5	Remove the ; at the end of Read(num1) and compile again. Read the error message, then put the ; back.

A Valid or not valid?

8 marks

Tick one box for each identifier.

Identifier	Valid	Not valid
Student_Name	☐	☐
2ndTerm	☐	☐
Total Marks	☐	☐
Num1	☐	☐
avg-mark	☐	☐
begin	☐	☐
FirstName	☐	☐
Marks$	☐	☐

B Which data type?

8 marks

Tick the data type of each value.

Value	Integer	Real	Boolean	Char	String
46	☐	☐	☐	☐	☐
25.68	☐	☐	☐	☐	☐
'k'	☐	☐	☐	☐	☐
'Sri Lanka'	☐	☐	☐	☐	☐
True	☐	☐	☐	☐	☐
-12	☐	☐	☐	☐	☐
'7'	☐	☐	☐	☐	☐
0.0	☐	☐	☐	☐	☐

"True AND True is True. Hard work AND practice is success."

- THE KING OF ICT

C Work out the results

8 marks

Write the result of each expression.

1. `17 DIV 5 =` .....
2. `17 MOD 5 =` .....
3. `15 / 4 =` .....
4. `4 + 10 MOD 3 =` .....
5. `2 + 3 * 4 =` .....
6. `18 / 3 * 2 =` .....
7. `9 > 4 =` .....
8. `6 < > 6 =` .....

D Find the reserved words

6 marks

Underline the six reserved words of Pascal.

```
program    Total    begin    num1    var    Sum    for    avg
then    Name    until    Marks
```

E Find the errors

5 marks

The program has five errors. Write the line number and the correct statement.

```
1  program add Num (input,output);
2  var
3      num1,num2,total : integer
4  Begin
5      Writeln("Enter First Number");
6      Read(num1);
7      Writeln('Enter Second Number');
8      Read(num2);
9      total = num1 + num2;
10     writeln('Total is ', total);
11 End
```

Line	The correct statement

WRITING SPACE 10 - Data types

write what your teacher tells you

"Errors are not failures. The compiler is your friend: it shows you where to look."

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REVISION

Unit 1 Part 1 in One Page

Read this the night before the test

Topic	The key facts
Analyzing a problem	Input = the raw materials; process = converting the input to the output, step by step; output = the result.
Alternative solutions	More than one solution to a problem. All of them = the solution space. Select the most appropriate one: (length + width) × 2.
Algorithm	A step-by-step procedure for solving a problem, with a plan.
Control structures	Sequence (strict order), selection (a condition decides), repetition (repeated until a condition is satisfied).
Flowchart symbols	Rounded box start / end, parallelogram input / output, rectangle process, diamond decision, arrows flow direction, circle connector.
Pseudo code	Simple English terms: BEGIN, END, INPUT / READ, DISPLAY / OUTPUT, CALCULATE, IF ... ENDIF, WHILE ... ENDWHILE, REPEAT ... UNTIL.
Converting	A loop in a flowchart becomes WHILE ... ENDWHILE; a decision becomes IF ... THEN ... ELSE ... ENDIF.
Identifier	A name for a variable, constant or program: starts with a letter; letters, numbers, _ ; no space, no reserved word.
Data types	Integer 46, Real 25.68, Boolean True / False, Char 'k', String 'ICT'.
var and const	A variable changes its value (var count : integer;); a constant does not (const max = 100;).
Operators	+ - * / DIV MOD; > >= < <= = < >; AND, OR, NOT. 20 DIV 6 = 3, 20 MOD 6 = 2.
Precedence	NOT first; then * / DIV MOD AND; then + - OR; then the comparisons.
A Pascal program	program name; var ...; Begin ... End. ; ends a statement, := assigns, { } is a comment.

★ TICK WHEN YOU CAN DO IT

- ☐ I can identify the input, the process and the output of a problem.
- ☐ I can find alternative solutions and choose the most suitable one.
- ☐ I can write an algorithm and name its control structures.
- ☐ I can draw a flowchart with the correct symbols.
- ☐ I can write a pseudo code and convert a flowchart into one.
- ☐ I can write valid identifiers and choose the right data type.
- ☐ I can work out expressions with DIV, MOD, AND, OR and NOT.

"Past O/L papers love algorithms, flowcharts and Pascal. Practise them all."

- THE KING OF ICT

WORD POWER

Unit 1 Part 1 Word List

Twenty-six words you must know

Word	What it means
Input	The raw materials (data) used to solve a problem.
Process	Converting the input to the output, step by step.
Output	The result obtained after solving a problem.
Alternative solutions	More than one solution to the same problem.
Solution space	All the solutions pertaining to a problem.
Algorithm	A step-by-step procedure for solving a problem.
Control structure	Sequence, selection or repetition in an algorithm.
Sequence	Steps carried out in a strict order from beginning to end.
Selection	Steps executed depending on whether a condition is satisfied.
Repetition	Steps repeated until a condition is satisfied.
Condition	A test whose answer is true or false (Yes or No).
Flowchart	A drawing of an algorithm with symbols joined by arrows.
Connector	The small circle of a flowchart that joins its parts.
Pseudo code	An algorithm written in simple English terms.
Variable	An identifier whose value changes while the program runs.
Constant	An identifier whose value does not change (const).
Identifier	A name for a variable, a constant or a program.
Reserved word	A word defined in Pascal, e.g. begin, end, var.
Data type	Integer, Real, Boolean, Char or String.
Integer	A plus or minus whole number: 0, 46, -12.
Real	A plus or minus decimal number: 0.0, 25.68.
Boolean	A value that is True or False.
Char / String	One character ('k') / a sequence of characters ('ICT').
Operator	A sign for a calculation, a comparison or logic: +, >, AND.
DIV / MOD	Whole-number division (20 DIV 6 = 3) / the remainder (20 MOD 6 = 2).
Assignment (:=)	Gives a value to a variable: total := num1 + num2;

"Learn the words of programming today. Speak to the computer tomorrow."

- THE KING OF ICT

"A king is not born. A king is made, one program at a time."

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END OF PART 1

Unit 1 Part 1 Test

Time: 45 minutes Total: 40 marks

Name _____ **Class** _____ **Marks** ☐

1 Choose the correct answer

8 marks

Underline your answer.

1. The raw materials used to solve a problem are called the
   (a) output (b) process (c) input (d) algorithm
2. A step-by-step procedure for solving a problem is
   (a) a flowchart symbol (b) an algorithm
   (c) an identifier (d) a constant
3. Marking the register till the last name is called is
   (a) sequence (b) selection (c) repetition (d) a constant
4. The flowchart symbol for a decision is a
   (a) rectangle (b) parallelogram (c) diamond (d) small circle
5. Which pseudo-code word shows an output?
   (a) INPUT (b) CALCULATE (c) DISPLAY (d) ENDIF
6. Which identifier is valid?
   (a) 9A (b) Last-name (c) Num1 (d) Average Marks
7. The value 'Sri Lanka' has the data type
   (a) Integer (b) Char (c) Boolean (d) String
8. The value of 20 MOD 6 is
   (a) 2 (b) 3 (c) 3.33 (d) 6

2 Fill in the blanks

8 marks

One word (or sign) in each blank.

1. All the solutions pertaining to a problem are called the ..... .
2. If the steps of an algorithm are carried out in a strict order, it is a ..... .
3. In a flowchart, a ..... is used for an input or an output.
4. An algorithm presented in simple English terms is called a ..... .
5. In Pascal, ... is used to declare a constant.
6. In Pascal, ... is the assignment operator.
7. The result of a comparison is always a ..... value.
8. When a value is assigned to a variable, the previous value is ..... .

3 Work out the results

8 marks

Write the result of each expression.

1. `7 + 3 * 2 =` .....
2. `25 DIV 4 =` .....
3. `25 MOD 4 =` .....
4. `9 / 2 =` .....
5. `18 / 3 * 2 =` .....
6. `(5 > 3) AND (2 > 4) =` .....
7. `(5 > 3) OR (2 > 4) =` .....
8. `NOT (6 MOD 2 = 0) =` .....

4 Answer these

16 marks

Write in the space given.

1. Identify the input, the process and the output of "finding the area of a circle". (3 marks)

2. Name the three control structures. Give one example of each. (3 marks)

3. Draw a flowchart: input the basic salary of an employee, add Rs. 5000 and output the new salary. (4 marks)

4. Look at the program. What is its output? What does := do? (3 marks)

```
var a, b : integer;
begin
    a := 17;
    b := a MOD 5;
    a := a DIV 5;
    writeln(a, ' ', b);
end.
```

5. Write two differences between a variable and a constant. (3 marks)

"Read every question twice. Then answer like a king."

- THE KING OF ICT